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Artificial Intelligence
Problems nr 3

1. Describe the tree

Nodes: Each node will be a different position robot can step on

Branches: Each movement that can be done from one node to the other

Tree Expansion: Starts on the starting position and follows valid moves between nodes

2. What algorithms could be used?

Dijkstra:

GBFS

A*

The best one would be A* because it ensures that the track is optimal

3. Cost Function

The cost function would be sum of all (energy cost of current cell * distance moved), so basically and since cost of white cells is 0.1 times distance and grey is 0.5 times the distance, it could be written down as:

Energy Cost = 0.1 * distance (white cells) + 0.5 * distance (grey cells)

And since we won't jump diagonally from one cell to the other and each cell has the size of 1x1 it can be written as

Energy Cost = 0.1 * white steps + 0.5 * grey steps

4. Heuristic functions

$$h_1(n) = |x_{\text{current}} - x_{\text{goal}}| * 0.1 + |y_{\text{current}} - y_{\text{goal}}| * 0.1$$

$$h_2(n) = (\sqrt{((x_{\text{current}} - x_{\text{goal}})^2 + (y_{\text{current}} - y_{\text{goal}})^2)}) * 0.1$$

which would be the distance from the goal in straight line and in Manhattan distance, and we assume all the tiles to be white ☺

5. Build the tree that A* algorithm will traverse from the initial position=9, to position=12, using the best heuristic. Write the sequence of lists that search algorithm will build

Step number	Now explored	Lists after move	Visited After
1.	([9, 6]; 0,1; 0,5)	([9, 11]; 0,5; 0,7), ([9, 6, 1]; 0,2; 0,7), ([9, 6, 5]; 0,2; 0,7)	9,6
2.	([9, 6, 5]; 0,2; 0,7) No way back	([9, 11]; 0,5; 0,7), ([9, 6, 1]; 0,2; 0,7)	9,6,5
3.	([9, 11]; 0,5; 0,7)	([9, 6, 1]; 0,2; 0,7), ([9, 11, 16]; 1; 1.3)	9,6,5,11
4.	([9, 6, 1]; 0,2; 0,7),	([9, 6, 1, 2]; 0,3; 0,7), ([9, 11, 16]; 1; 1.3)	9,6,5,11,1
5.	([9, 6, 1, 2]; 0,3; 0,7),	([9, 6, 1, 2, 3]; 0,4; 0,7), ([9, 11, 16]; 1; 1.3)	9,6,5,11,1,2
6.	([9, 6, 1, 2, 3]; 0,4; 0,7)	([9, 6, 1, 2, 3, 7]; 0,5; 0,7), ([9, 6, 1, 2, 3, 4]; 0,5; 0,9), ([9, 11, 16]; 1; 1.3)	9,6,5,11,1,2,3
7.	([9, 6, 1, 2, 3, 7]; 0,5; 0,7),	([9, 6, 1, 2, 3, 7, 8]; 1; 1,3), ([9, 6, 1, 2, 3, 7, 10]; 0,6; 0,7), ([9, 6, 1, 2, 3, 4]; 0,5; 0,9), ([9, 11, 16]; 1; 1.3)	9,6,5,11,1,2,3,7
8.	([9, 6, 1, 2, 3, 7, 10]; 0,6; 0,7),	([9, 6, 1, 2, 3, 7, 8]; 1; 1,3), ([9, 6, 1, 2, 3, 7, 10, 12]; 1,1; 1,1), ([9, 6, 1, 2, 3, 4]; 0,5; 0,9), ([9, 11, 16]; 1; 1.3)	9,6,5,11,1,2,3,7,10
9.	([9, 6, 1, 2, 3, 4]; 0,5; 0,9),	([9, 6, 1, 2, 3, 7, 8]; 1; 1,3), ([9, 6, 1, 2, 3, 7, 10, 12]; 1,1; 1,1), ([9, 6, 1, 2, 3, 4, 8]; 1; 1,3), ([9, 11, 16]; 1; 1.3)	9,6,5,11,1,2,3,7,10,4
10.	([9, 6, 1, 2, 3, 7, 10, 12]; 1,1; 1,1),	THE END	

6. Now, assume that the robot can make movements in diagonal. Write an example of node where the previous heuristic overestimate the cost of the energy.

the Manhattan heuristic would overestimate in any situation where a diagonal move is possible, e.g. way from 13 to 19, it would estimate $(1 + 1) * 0.1 = 0.2$, but the real distance would be $\sqrt{2} * 0.1 \sim 0.141$.

- 7. What happens if we use GBFS instead of A*?, compare the results obtained by the two algorithms (in the case that only horizontal and vertical movements are allowed)**

In Greedy Best-First Search, the algorithm selected the next node based purely on the heuristic value, without considering the actual path cost. As a result, the exploration prioritizes nodes that seem closest to the goal according to the heuristic, potentially leading to suboptimal paths.

So, the GBFS would go with (9-11-16-17-18-12) instead of (9-6-1-2-3-7-10-12), which is the better route.